KAIZEN MATHS

Lesson starter

Unit Conversions and Compound Measures

Applications

Name _____

S1

Convert 14°F to °C.

S2

Convert 100 km/h to mph.

S3

Convert 2700 kg/m^3 to g/cm^3.

S4

Given A = 200, B = 4A, and C = 5B + 25. Find C.

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NUM-70D7-L2 · S

Show your working

KAIZEN MATHS

Exit ticket

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E1

Convert 20°C to K.

E2

Convert 7.2 m/s to km/h.

E3

Convert 12 cm/s to m/s.

Show your working

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NUM-70D7-L2 · E

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NUM-70D7-L2 · E

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Teacher answers & working

Unit Conversions and Compound Measures · Applications

NUM-70D7-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

Convert 14°F to °C.

Answer:

-10°C

Working:

$C = 5(F - 32)/9$

Substitute F = 14

$C = 5(14 - 32)/9$

$C = -10°C$

S2

Convert 100 km/h to mph.

Answer:

62.15 mph

Working:

1 mile = 1.609 km

Convert back by dividing by 1.609.

$100 \div 1.609 = 62.15$ mph

S3

Convert 2700 kg/m³ to g/cm³.

Answer:

2.7 g/cm³

Working:

$1\ kg/m^3 = 0.001\ g/cm^3$

$2700 \times 0.001 = 2.7\ g/cm^3$

S4

Given A = 200, B = 4A, and C = 5B + 25. Find C.

Answer:

$C = 4025$

Working:

First find B.

$B = 4 \times 200 = 800$

Now substitute B into the expression for C.

$C = 5 \times 800 + 25 = 4025$

Teacher answers & working

Unit Conversions and Compound Measures · Applications

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E1

Convert 20°C to K.

Answer:

293.15 K

Working:

$K = C + 273.15$

$K = 20 + 273.15 = 293.15$ K

E2

Convert 7.2 m/s to km/h.

Answer:

25.92 km/h

Working:

1 m/s $= 3.6$ km/h

$7.2 \times 3.6 = 25.92$ km/h

E3

Convert 12 cm/s to m/s.

Answer:

0.12 m/s

Working:

1 cm $= 1/100$ m

$12 \div 100 = 0.12$ m/s